

K-Coverage for Irregular Targets in Wireless Visual Sensor Networks Deployed in Region of Interest Containing Obstacles^{*}

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ABSTRACT

In wireless visual sensor networks (WVSNs), coverage estimation is essential for optimizing network energy consumption and reducing deployment costs. In recent years, the majority of literature primarily focuses on small-scale networks, and typically assumes that visual sensors are either predetermined or randomly deployed in obstacle-free region of interest (RoI). However, in real-world scenarios, these sensors are frequently deployed in challenging-to-access environments that contain irregular obstacles. Moreover, the targets in these environments exhibit considerable diversity in shape and size. To address these challenges, this paper presents an issue of irregular targets K coverage with obstacles (K -ITCO) in WVSNs deployed in irregular RoI. In addition, we account for the probability that targets are unobstructed and consider the boundary effects. We derive two K -ITCO estimation models and their respective solving methods (K -ITCO-BE and K -ITCO-nBE) that predict a specified coverage ratio with a high probability. To validate the effectiveness and accuracy of our proposed models, multiple sets of simulations are designed to analyze the numerical and simulation results. Our findings demonstrate a high degree of consistency with the simulation outcomes, and indicate that the model can effectively estimate K -ITCO ratio.

1. Introduction

With the rapid advancements in wireless visual sensor and embedded processor technologies, wireless visual sensor networks (WVSNs) have emerged as a fundamental component of the Internet of Things (IoT), undergoing substantial growth in recent years. Unlike traditional wireless sensor networks (WSNs), WVSNs comprise numerous visual sensors (sensors) equipped with sector-shaped sensing regions that enable flexible adjustments in sensing orientations. These networks are capable of capturing comprehensive environmental data in the form of images and videos, significantly enhancing the diversity and utility of the collected data. In small-scale applications such as indoor security monitoring, bridge monitoring, and urban traffic surveillance[1][2], sensor positions and sensing orientations are typically predetermined to ensure efficient quality of coverage (QoC). This preset deployment strategy ensures optimal coverage and data collection efficiency within specific monitoring area. However, in some specialized application scenarios, such as wildlife conservation, intelligent battlefield surveillance, forest fire surveillance, and smart agricultural monitoring[3][4][5][6], the regions of interest (RoI) are often challenging to access, making it impractical to manually deploy sensors with precise positions and sensing orientations. Consequently, in these cases, sensors are generally dispersed randomly by aircraft to facilitate rapid and effective monitoring. In conclusion, WVSNs have demonstrated considerable potential and application value across various fields, attracting significant attention from scholars.

Generally, the region of interest monitored by visual sensor network typically exhibits an irregular shape, rather than simple rectangle or square, and often contains numerous irregular obstacles, such as trees, hills, rocks, buildings,

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vehicles, and walls[7][8]. These obstacles frequently obstruct targets, leading to a reduction in QoC, as illustrated in Fig.1(a). Additionally, the shapes and sizes of targets located within RoI are often diverse and unpredictable. For example, if the targets are animals, plants, people or vehicles in grassland, desert or urban environments, they are represented by the entities themselves. Alternatively, if the monitoring target is an area where animals or people frequently appear, the area with a highest probability of observation should be considered as the target[9]. Despite these complexities, most of existing literature simplifies target to a single point model, which inadequately reflects real-world scenarios. Furthermore, these irregular shaped targets may obstruct one another, as shown in Fig.1(b), where a target vehicle obscures a people (red color). The randomness in positions and shapes of the RoI, obstacles and targets further compound the complexity of target coverage estimation.

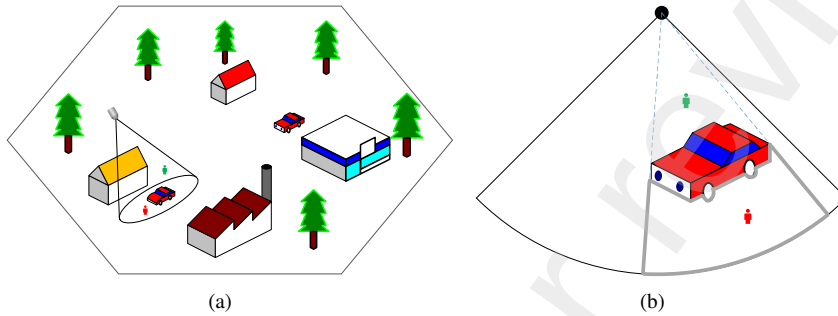


Figure 1: Illustration of real-world scenario and visual occlusion. (a) deployment in irregular RoI, the car and the red color people are obstructed by the house with a yellow roof, (b) visual occlusion, the red color people is obstructed by the car, and green color people is unobstructed.

Additionally, in the context of the large-scale application scenarios explored in this paper, visual sensors are randomly deployed in the RoI. Consequently, some sensors may be positioned near the boundary region, resulting in boundary effects, which lead to the reduction of the actual sensing region of each sensor compared to its theoretical value. Therefore, this paper primarily focuses on the following key issue: **How can target coverage ratio be accurately estimated before initial deployment, fully accounting for the shapes and sizes of both targets and obstacles, while also considering the influence of boundary effects and an irregular RoI?** Solving this issue is expected to effectively reduce deployment costs and improve the deployment efficiency. To the best of our knowledge, no existing research directly addresses this specific issue.

The primary contributions of this study can be summarized as follows:

- Considering the shapes and sizes of targets and obstacles, we first present novel models for irregular target and obstacle. Subsequently, we incorporate the irregular characteristics of the RoI. Additionally, we introduce the system model and problem formulation.
- Based on the target and obstacle models, we propose a novel target coverage metric—irregular target coverage with obstacles (*K*-ITCO). Subsequently, we introduce the concepts of the detection region (DR) and the effective detection region (EDR), and derive the expected value of EDR of a sensor. Furthermore, we deduce the probability that targets are unobstructed. Additionally, we further derive the *K*-ITCO estimation model accounting for boundary effects, along with its algorithm (*K*-ITCO-BE). Finally, we present the relevant *K*-ITCO estimation model that neglects boundary effects and its corresponding algorithm (*K*-ITCO-noBE).
- A series of simulations are conducted to validate the accuracy of numerical results obtained by our proposed algorithms.

The remainder of this paper is organized as follows: Section II provides a brief overview of notable contributions in related work. Section III presents the relevant models and problem formulation, along with definitions. In section IV, we provides the derivation of the expected value of EDR. Section V introduces the derivation of the *K*-ITCO estimation models. In section VI, we conducts multiple simulations to evaluate the accuracy of the numerical results calculated by *K*-ITCO-BE and *K*-ITCO-noBE. Finally, Section VII concludes the paper and discusses potential directions for future research.

2. Related Work

With the growing prevalence of the Internet of Things, a substantial amount of work has been dedicated to addressing the coverage issues in WVSNs. In this section, we briefly summarize the notable contributions related to these coverage challenges.

Concerning the issue of coverage estimation in WVSNs, in [10], the authors investigated a novel target detection model and presented a closed-form solution for coverage estimation that considers the shape of target, but did not account for the presence of obstacles. Based on the results of this work, Liu et al [11] assumed that an irregular field of interest (FoI) with the presence of irregular obstacles, while also considering the boundary effects. Then, they presented a new occlusion k -area coverage issue and deduced a critical sensor density model to predict k -area coverage ratio in heterogeneous WVSNs. However, they did not consider the irregular target coverage issue and the occlusion between targets. Additionally, the authors in [12] introduced the concept of partial target occlusion coverage for WVSNs deployed in a square area containing obstacles, and derived a critical sensor density to achieve an expected coverage ratio with a high probability, but did not consider the shape of target and the shape of FoI. Aiming for large-scale application, the work in [13] derived an analytical expression for stochastic coverage based on both sensor and network parameters in WVSNs. In [14], disregarding obstacles, the authors introduced fixed-orientation directional sensor networks and analytically derived critical sensor density for both area coverage and connectivity. They further explored critical sensor density for both area coverage and connectivity in adjustable-orientation directional sensor networks [15]. Kang et al [16] derived the critical density for non-orientation directional sensor networks and proposed an approach to compute the critical density, but ignoring the presence of obstacles and boundary effects. The authors in [17] introduced the concept of equivalent sensing radius (ESR) and derive ESR under three different mobility models—1-dimensional and 2-dimensional and random rotating model—then derived critical conditions required to achieve almost sure coverage ratio 1 in WVSNs. However, they did not consider the shapes of targets, the presence of obstacles and partial coverage. Wang et al [18] proposed a novel model called full-view coverage in WVSNs and derived a sufficient condition on the sensor density needed to achieve full-view coverage. Additionally, they introduced full-view barrier coverage problem and a corresponding detection method, though the boundary effects and obstacles are not considered. By taking the boundary effects into account, Liu et al [19] formulated a sensor parameters prediction model to achieve a desired full-view area coverage ratio in heterogeneous WVSNs deployed in an irregular field of interest.

We now briefly summarize the notable contributions related to coverage optimization problem in WVSNs. Shi et al [20] investigated the issue of minimum number of sensors needed to achieve full-view point coverage ratio of 1, and proposed two approximation algorithms—iterative screening algorithm (ISA) and improved ISA—to solve that problem. The authors in [21] focused on moving targets and introduced the issue of maximizing full-view target coverage in WVSNs. Then, they proposed two algorithms—maximum group set coverage (MGSC) and MGSC-composed—to tackle this issue. Considering real-world spatiotemporal constraints such as buildings and pedestrian dynamics, Li [22] proposed a quantum annealing (QA)-based full-view area coverage problem (FVACP) optimization framework. This approach addressed spatial constraints by optimizing deployment points and tackling temporal constraints by optimizing sensing orientations. In [23], the authors investigated joint optimization of sensors nodes deployment, edge nodes deployment, data routing and data offloading to minimize the number of sensors, edge nodes and network lifetime, formulating this as a mixed-integer nonlinear program and presenting a multi-objective differential evolution algorithm to solve it. The authors in [24] studied maximum weighted target coverage in rechargeable visual sensor networks charged by UAVs in 3D space, and formulated this issue as a mixed integer nonlinear program. Then, this work presented a greedy heuristic and a particle swarm optimizer to solve it. Chen et al [25] studied k -barrier coverage issue in hybrid visual sensor networks and presented a strong k -barrier coverage enhancement scheme to transfer this issue into barrier gaps repairing problem in k turns, aiming to minimize the number of mobile sensors and the total moving distance. In [26], the authors propose a novel 2D target coverage model based on sensor parameters, the target positions and sensor positions, and a heuristic approach for sensors selection and focal adjustment is presented to optimize coverage quality while maintaining high image quality.

In summary, most prior studies either assume that the deployment area is a regular region without obstacles or that the target is modeled as a point. Moreover, no existing research has investigated the K -coverage problem for irregular targets in WVSNs deployed in irregular region of interest containing obstacles.

3. Preliminaries

3.1. Sensing Model

As illustrated in Fig.2, the sensing model of a visual sensor (sensor) is typically represented by a 4-tuple $\langle s, r, \vec{f}, \theta \rangle$. Where s denotes the sensor along with its location, r represents the sensing radius, $\vec{f}$ indicates the sensing orientation, and θ refers to the field of view (FoV) angle. Based on this model, the sensing region of the visual sensor can be described geometrically as a sector.

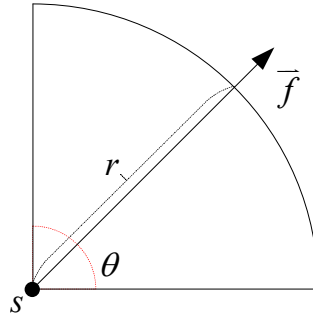


Figure 2: Sensing model of a visual sensor.

3.2. Target Model

Most of the existing literature assumes that the shape and size of a target are negligible, which is inadequate for real-world scenarios. For instance, if the targets are animals, plants or people in grassland, desert or urban environments, the targets are entities themselves. Alternatively, If the target is an area where animals or people frequently appear, the target should be considered as the area with the highest probability of observation. In either case, the target's shape can be described as an irregular polygon. According to literature [27], any irregular polygon could be decomposed into multiple convex polygons. Hence, in this paper, we take irregular convex target into consideration. As depicted in Fig.3, the target model is represented as a 5-tuple $\langle t, \mathbf{tv}, t_c, tr, t\beta \rangle$, where t is the target itself and its location, $\mathbf{tv} = \{tv_i\}_{i=1 \dots n_t}$ denotes the vertices of the target, and t_c refers to its centroid. Furthermore, we define tr as the target's minimum bounding rectangle calculated by the algorithm described in [28], which its length and width are represented as tl and tw , respectively. The tr 's rotation angle relative to the X-axis is denoted by $t\beta$.

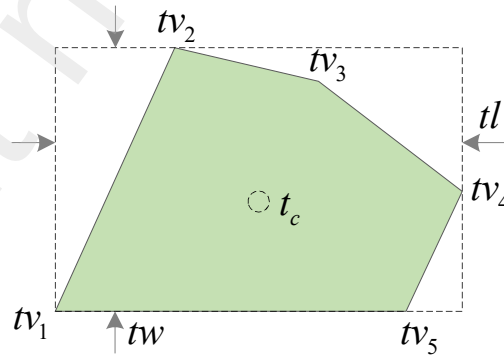


Figure 3: Model of an irregular target. The target is depicted as the green convex polygon.

3.3. Obstacle Model

Similar to the target model, and in accordance with literature [27], we take convex obstacles into consideration. As shown in Fig.4, the obstacle model is also represented as a 5-tuple $\langle o, \mathbf{ov}, o_c, or, o\beta \rangle$, where o denotes the obstacle

and its location, $\mathbf{ov} = \{ov_i\}_{i=1 \dots n_o}$ is the vertices of the obstacle, and o_c refers to its centroid. In addition, let or be the minimum bounding rectangle of the obstacle, and the variables ol and ow indicate the length and width of or , respectively. And $o\beta$ denotes the rotation angle of or relative to the X -axis.

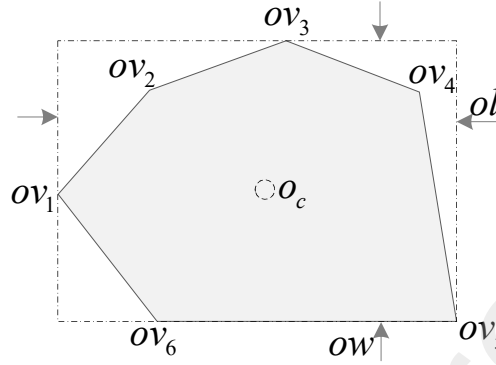


Figure 4: Model of irregular obstacle. The obstacle is represented by the Gray convex polygon.

3.4. System Model and Problem Formulation

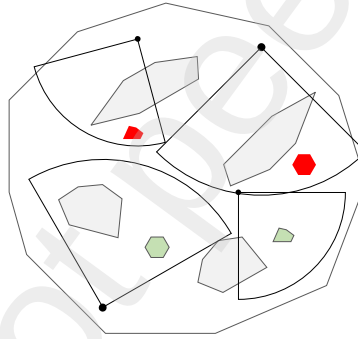


Figure 5: Illustration of system model. Multiple types sensors are randomly scattered in an irregular RoI. The obstacles are represented by the gray convex polygons. The targets being obstructed are depicted as the red convex polygons, while the targets being unobstructed are denoted by the green convex polygons.

To formulate the system model studied in this paper, as depicted in Fig.5, we make the following assumptions:

A1: N visual sensors, M irregular targets and W irregular obstacles are randomly and independently located in an irregular RoI denoted as Ω .

A2: The sensors are categorized into u distinct types, all sensors in identical $Type_i$ possesses same characteristics, including sensor scale, sensing radius and FoV. However, the relevant parameters for sensors from different types vary. Similarly, we divide obstacles into v groups, the i^{th} group is represented as GO_i , and targets into w groups, the i^{th} group is denoted as GT_i . Obviously, obstacles or targets in the same group have identical shape and size.

Mathematical Description: Let $N = \sum_{i=1}^u N_i$, where N_i denotes the number of sensors in $Type_i$. For sensors in $Type_i$ and $Type_j$, if $i \neq j$, then $r_i \neq r_j$, $N_i \neq N_j$ and $\theta_i \neq \theta_j$. Similarly, let $M = \sum_{i=1}^v M_i$, where M_i represents the number of targets in GT_i . For targets belonging to GT_i and GT_j , if $i \neq j$, then $t_i \neq t_j$, $tc_i \neq tc_j$ and $tr_i \neq tr_j$. Likewise, let $W = \sum_{i=1}^w W_i$, where W_i denotes the number of obstacles in GO_i . For obstacles in different groups (GO_i and GO_j), if $i \neq j$, then $o_i \neq o_j$, $oc_i \neq oc_j$ and $or_i \neq or_j$ will hold.

Given the system model, the K -ITCO estimation issue is formulated as follows:

Problem: Given an initial random deployment of a predefined number of visual sensors in an irregular RoI containing obstacles. Meanwhile, considering the shape and size of the targets and the boundary effects, how to

estimate the *K*-ITCO ration with a high probability? Alternately, if a desired *K*-ITCO ratio is specified, how to predict the number of visual sensors needed to be randomly deployed in the RoI?

3.5. Definitions

3.5.1. Boundary Zone (BZ) and Center Region (CR)

From the Fig.6, given a sensing radius r , we observed that the RoI can be categorized into two distinct parts: the gray part is the boundary zone denoted as *BZ*, and the remaining central part is the center region denoted as *CR*. Meanwhile, the terms *BZ* and *CR* can be used to represent their respective areas.

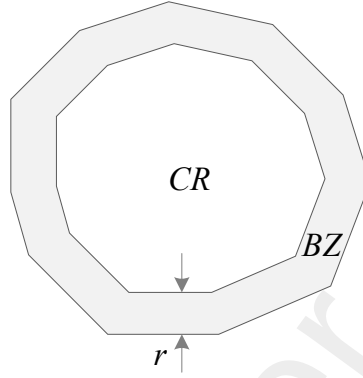


Figure 6: Illustration of boundary zone (*BZ*) and central region (*CR*).

3.5.2. Detection Region (DR)

For a visual sensor, its detection region is defined as the entire disk that can be monitored by rotating its sensing orientation. Here, let ϕ represent the detection region (DR) of the sensor, it is obvious that the area of ϕ is given by πr^2 , where r is the sensor's sensing radius.

3.5.3. Effective Detection Region (EDR)

As illustrated in Fig.7, the DR ϕ of a sensor would intersect with the RoI Ω , the intersection area is referred to as the effective detection region denoted as $\Phi = \phi \cap \Omega$. Obviously, its area is generally less than that of ϕ .

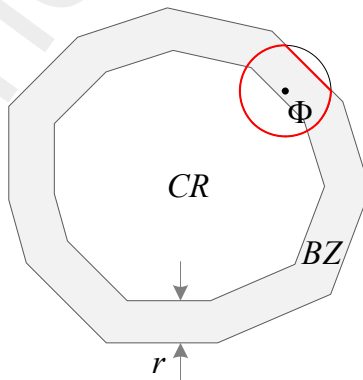


Figure 7: Illustration of effective detection region (EDR). The EDR is depicted as the red-lined region.

3.5.4. Irregular Target Coverage with Obstacles (ITCO)

Concerning an irregular target t and a RoI, the shapes and sizes of obstacles and other targets have a negative impact on the quality of coverage (QoC). Thus, the conventional coverage model of target cannot directly address this issue.

As depicted in Fig.8, the target t is considered to be ITCO-covered if and only if the following three mathematical conditions are satisfied:

- (1) $dist(s, t_c) \leq r$, where r denotes the sensing radius of a sensor s and t_c is the centroid of the target t .
- (2) $\alpha(\vec{f}, st_c) \leq \theta/2$, where $\vec{f}$ represents the sensor's sensing orientation, and θ is its FoV angle.
- (3) For any obstacle o_i or other target t_i where $t_i \neq t$, if $\exists p_j^{o_i}(x_j^{o_i}, y_j^{o_i}) = st \cap edgesof(o_i)$ or $\exists p_j^{t_i}(x_j^{t_i}, y_j^{t_i}) = st \cap edgesof(t_i)$, Then the following must hold:
 $(x_s - x_j^{o_i})(x_t - x_j^{o_i}) > 0$ and $(x_s - x_j^{t_i})(x_t - x_j^{t_i}) > 0$
 where $j \in \{1, 2\}$. The terms x_s and x_t represent the X -axis coordinates of s and t , respectively.

The Conditions 1 and 2 guarantee that the centroid t_c of the target t lies within the sensing region of the sensor s , while condition 3 implies that the target t is not obstructed by obstacles or other targets.

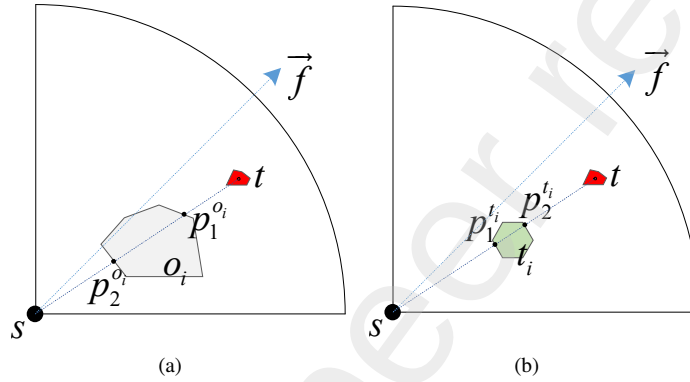


Figure 8: Illustration of a target being obstructed. The target t is represented by the red convex polygon. (a) Being obstructed by obstacle o_i , (b) Being obstructed by other target t_i .

3.5.5. K-ITCO-Covered

A target t is said to be K -ITCO-covered if it can be ITCO-covered by at least K visual sensors, where K represents the coverage degree.

3.6. Main Symbols

To better describe the problem, the main symbols used in this paper are summarized in Table 1.

4. Derivation of the Expected Value of EDR

Aiming to the system model we studied in this paper, we derive an expression for the expected value of EDR for a randomly selected sensor s lying in the irregular RoI.

In accordance with the position of the sensor s , we take two cases into consideration: 1) if the sensor lies in the central region CR , let $E(\Phi_c)$ represent the expected value of its EDR. Clearly, $E(\Phi_c) = \phi = \pi r^2$, where ϕ denotes its DR; and 2) if the sensor located in the boundary zone BZ , let $E(\Phi_b)$ denote the expected value of its EDR. It is obvious that $E(\Phi_b)$ is smaller than its DR ϕ . Afterwards, let $E(\Phi)$ denote the expected value of EDR of the sensor in the RoI, it is apparent that $E(\Phi)$ depends on the sensor's location. Due to the fact that the sensor is either randomly distributed in BZ or CR , the expression for $E(\Phi)$ can be calculated using the following formula:

$$E(\Phi) = \frac{CR}{\Omega} E(\Phi_c) + \frac{BZ}{\Omega} E(\Phi_b) \quad (1)$$

In the remainder of this section, we derive the formulas for calculating the areas of RoI Ω , central region CR and boundary zone BZ . Subsequently, we approximate the BZ as an equivalent rectangle zone to estimate the expression for $E(\Phi_b)$.

Table 1
Description of Symbols

Symbol	Description
Ω	Irregular RoI
u	Number of sensor types
N_i	Number of sensors in $Type_i$
$N = \sum_{i=1}^u N_i$	Total number of sensors in the RoI
λ_i	Density of $Type_i$ sensor
r_i	Sensing radius of $Type_i$ sensor
θ_i	FoV angle of $Type_i$ sensor
v	Number of target groups
M_i	Number of targets in GT_i
$M = \sum_{i=1}^v M_i$	Number of targets in the RoI
μ_i	Density of GT_i target
w	Number of obstacle groups
W_i	Number of obstacles in GO_i
$W = \sum_{i=1}^w W_i$	Number of obstacles in the RoI
η_i	Density of GO_i obstacle

4.1. Areas of RoI, CR and BZ

We assumed that the irregular RoI Ω has n vertices, represented as $a_i(x_i^a, y_i^a)$ where $0 \leq i \leq n$. Accordingly, the corresponding central region CR also has n vertices, denoted as $c_i(x_i^c, y_i^c)$. As illustrated in Fig.9, let $\vec{v}_1$ and $\vec{v}_2$ denote the unit vectors of $\overrightarrow{a_i a_{i-1}}$ and $\overrightarrow{a_i a_{i+1}}$, respectively. In accordance with **Definition 1** and the principles of spatial analytic geometry, it is concluded that the coordinates of the vertex $c_i(x_i^c, y_i^c)$ can be determined using the following formulas:

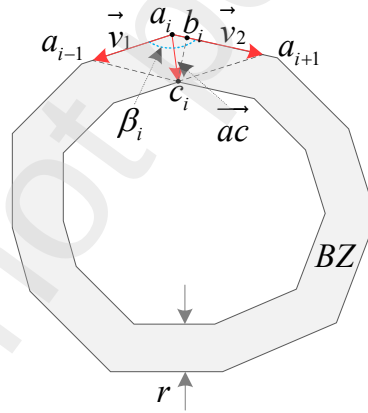


Figure 9: Illustration of vertices of the RoI and the CR. It also shows geometric relations of the relevant vectors.

$$\begin{aligned} x_i^c &= x_i^a + \frac{r}{\sin(\beta_i/2)} \frac{x^{ac}}{\text{Normalize}(\vec{ac})} \\ y_i^c &= y_i^a + \frac{r}{\sin(\beta_i/2)} \frac{y^{ac}}{\text{Normalize}(\vec{ac})} \end{aligned} \quad (2)$$

Where r denotes the sensing radius of the randomly selected sensor s , β_i represents the angle between the vectors $\overrightarrow{a_i a_{i-1}}$ and $\overrightarrow{a_i a_{i+1}}$. Thus, we get that $\sin(\beta_i/2) = \sqrt{(1 - \vec{v}_1 \cdot \vec{v}_2)}/2$, here $\vec{v}_1 \cdot \vec{v}_2$ denotes the dot product of the vectors. Moreover, the coordinates of the vector $\vec{ac} = \vec{v}_1 + \vec{v}_2$ are represented by x^{ac} and y^{ac} .

Subsequently, based on the coordinates of the vertices, the areas of RoI Ω , central region CR and boundary zone BZ are respectively expressed as follows:

$$\Omega = \frac{\left| \sum_{i=1}^n \left(x_i^a y_{(i+1) \bmod n}^a - x_{(i+1) \bmod n}^a y_i^a \right) \right|}{2} \quad (3)$$

$$CR = \frac{\left| \sum_{i=1}^n \left(x_i^c y_{(i+1) \bmod n}^c - x_{(i+1) \bmod n}^c y_i^c \right) \right|}{2} \quad (4)$$

$$BZ = \Omega - CR \quad (5)$$

4.2. Expected Value $E(\Phi_b)$ of EDR for a Sensor in the Boundary Zone (BZ)

A randomly selected sensor s in the boundary zone BZ would experience boundary effects, causing its actual sensing region to be smaller than its theoretical value. Taking the boundary effects into account, we derive the expected value of its EDR, denoted as $E(\Phi_b)$. Before proceeding with derivation, we present the following **Lemma 1**.

Lemma 1: The boundary zone BZ can be approximated as a rectangle zone RZ which its width is r and length is L , where L is given by:

$$L = \sum_{i=1}^n \left(d_i - \frac{r}{2 \tan(\beta_i/2)} - \frac{r}{2 \tan(\beta_{(i+1) \bmod n}/2)} \right).$$

Here d_i denotes the Euler distance between vertices a_i and a_{i+1} .

Proof: As illustrated in Fig.9, it is clear that $a_i b_i = \frac{r}{\tan(\beta_i/2)}$. Thus, we get that $S_{\triangle a_i b_i c_i} = \frac{r^2}{2 \tan(\beta_i/2)}$. Using principles of spatial analytic geometry, we can conclude the following formula:

$$\begin{aligned} BZ &= \sum_{i=1}^n \left(r d_i - \frac{r^2}{2 \tan(\beta_i/2)} - \frac{r^2}{2 \tan(\beta_{(i+1) \bmod n}/2)} \right) \\ &= r \times \sum_{i=1}^n \left(d_i - \frac{r}{2 \tan(\beta_i/2)} - \frac{r}{2 \tan(\beta_{(i+1) \bmod n}/2)} \right) \end{aligned}$$

Obviously, the **Lemma 1** is proven.

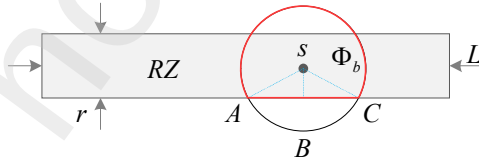


Figure 10: Illustration of the EDR of a sensor approximately lies in the RZ.

Based on **Lemma 1**, the rectangle zone RZ can be used as an approximation for the boundary zone BZ . Accordingly, the sensor s located in the BZ can be approximated as residing within the RZ . As depicted in Fig.10, let (x, y) denote the coordinates the sensor s . Hence, its EDR denoted as Φ_b can be estimated by:

$$\Phi_b = \pi r^2 - (S_{\widehat{sABC}} - S_{\triangle sAC}) \quad (6)$$

$$\text{Where } S_{\widehat{sABC}} = r^2 \arccos\left(\frac{y}{r}\right) \text{ and } S_{\triangle sAC} = \frac{r^2 \sin\left(2 \arccos\left(\frac{y}{r}\right)\right)}{2}.$$

It is evident that the value of Φ_b depends on the sensor's coordinates. Therefore, expected value $E(\Phi_b)$ of the EDR for the sensor s can be obtained by simply integrating Φ_b over RZ , which can be calculated using the following formula:

$$E(\Phi_b) = \iint \Phi_b d(RZ) = \int_0^r dy \int_0^L \Phi_b dx \quad (7)$$

Afterwards, by substituting Eq.3 4 5 7 into Eq.1, we obtain the exact expression for calculating the expected value of the EDR for the randomly selected sensor s .

5. Derivation of the K-ITCO Estimation Model

In this section, for a randomly selected irregular target t , we consider the probability that it is not obstructed by obstacles or other targets. Subsequently, the K -ITCO estimation model is derived to achieve a specified coverage ratio with a high probability. Moreover, some propositions are proposed. Before proceeding with the derivation, the following lemma is presented.

Lemma 2. Consider an area R containing N points, and let R' be a random subarea of R . Additionally, it is assumed that X be a random variable representing the number of points in R' . Obviously, the event $\{X = K\}$ signifies that exactly K points are located within the subarea R' . Thus, the probability of this event can be expressed as:

$$P(X = K) \approx e^{-\lambda R'} \frac{(\lambda R')^K}{K!} \quad (8)$$

Proof: Using F be a random variable denoting whether a point is located in the subarea R' , where F takes the value of 1 if the point is in R' and 0 otherwise. The event $\{F = 1\}$ indicates that a point lies within R' , and its probability is given by $P(F = 1) = \frac{R'}{R}$. According to the Binomial Distribution, the probability of the event $\{X = K\}$, representing exactly K points in R' , can be expressed as:

$$P(X = K) = \binom{N}{K} (P(F = 1))^K (1 - P(F = 1))^{N-K}$$

Assume that $\lambda = \frac{N}{R}$, we can derive the following proof process.

$$\begin{aligned} & \lim_{N \rightarrow \infty} P(X = K) \\ &= \lim_{N \rightarrow \infty} \binom{N}{K} (P(F = 1))^K (1 - P(F = 1))^{N-K} \\ &= \lim_{N \rightarrow \infty} \frac{N!}{(N-K)!K!} \left(\frac{R'}{R}\right)^K \left(1 - \frac{R'}{R}\right)^{N-K} \\ &= \lim_{N \rightarrow \infty} \frac{N!}{(N-K)!K!} \left(\frac{\lambda R'}{N}\right)^K \left(1 - \frac{\lambda R'}{N}\right)^N \left(1 - \frac{\lambda R'}{N}\right)^{-K} \\ &= e^{-\lambda R'} \frac{(\lambda R')^K}{K!} \end{aligned}$$

Thus, the Lemma 2 is proven.

5.1. Probability that a Target is Unobstructed

Based on the system model we studied, given a randomly chosen irregular target t and a visual sensor s , it is obvious that obstacles and other targets potentially generate visual occlusion zones obstructing the line of sight from the sensor s to the target t . In this subsection, we assume that the centroid of t is consistently located within the sensing region of the sensor s . Subsequently, we derive the probability that the target is not obstructed by obstacles or other targets.

As illustrated in Fig.11(a), let ZO_i denote the visual occlusion zone formed by an obstacle originating from GO_i . The area of ZO_i depends on two factors: 1) the distance do_i from the center point of the minimum bounding rectangle or_i to the line segment st , where $\frac{ow_i}{2} \leq do_i \leq \frac{\sqrt{ol_i^2 + ow_i^2}}{2}$; and 2) the distance x between the sensor s and the target t , where $0 \leq x \leq r$. Thus, the expected area of ZO_i can be calculated using the following formula:

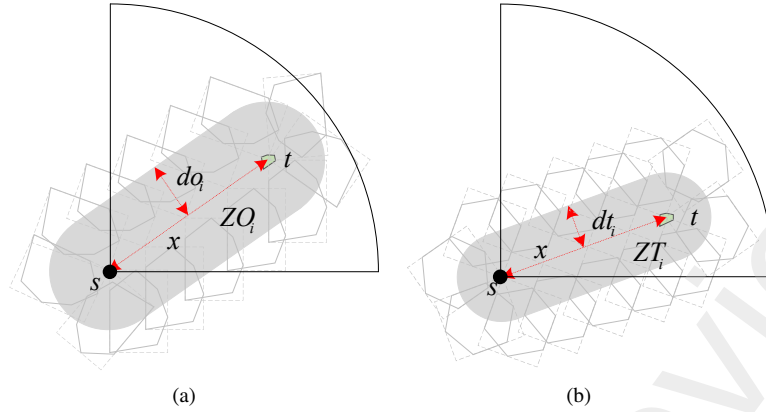


Figure 11: Visual occlusion zones are constructed by obstacles or other targets. (a) Visual occlusion zone ZO_i is formed by obstacles from GO_i , (b) Visual occlusion zone ZT_i is formed by other targets from GT_i .

$$E(ZO_i) = \pi E(dO_i)^2 + 2E(dO_i)x \quad (9)$$

Where $E(dO_i) = \frac{ow_i + \sqrt{ol_i^2 + ow_i^2}}{4}$.

As shown in Fig.11(b), let ZT_i denote the visual occlusion zone generated by a target originating from GT_i which does not belong to the same group as the target t . Following similar deduction as above, we can readily conclude that the expected area of ZT_i can be given by:

$$E(ZT_i) = \pi E(dT_i)^2 + 2E(dT_i)x \quad (10)$$

In this context, $E(dT_i) = \frac{tw_i + \sqrt{tl_i^2 + tw_i^2}}{4}$, where dT_i denotes the distance from the center point of the minimum bounding rectangle tr_i to the line segment st .

To ensure that the selected target t is not obstructed by obstacles from GO_i or other targets from GT_i , the centroid points of these obstacles and targets must not be located within the zones $E(ZO_i)$ and $E(ZT_i)$, respectively. Let Y_{ZO_i} be a random variable indicating the number of obstacles located in $E(ZO_i)$, and let Y_{ZT_i} be a random variable denoting the number of targets lying in $E(ZT_i)$. According to **Lemma 2**, the probabilities of the events $\{Y_{ZO_i} = 0\}$ and $\{Y_{ZT_i} = 0\}$ are given by:

$$P(Y_{ZO_i} = 0) = e^{-\eta_i E(ZO_i)} \quad (11)$$

$$P(Y_{ZT_i} = 0) = e^{-\mu_i E(ZT_i)} \quad (12)$$

Where $\eta_i = \frac{W_i}{\Omega}$ represents the density of obstacles from GO_i per unit area, and $\mu_i = \frac{M_i}{\Omega}$ denotes the density of targets from GT_i per unit area.

Subsequently, we assume that random variable $Y = \sum_{i=1}^w Y_{ZO_i} + \sum_{i=1}^v Y_{ZT_i}$ represents the total number of obstacles from different groups falling into their respective visual occlusion zones, as well as the total number of targets from different groups lying in their corresponding visual occlusion zones. It can be inferred that the event $\{Y = 0\}$ signifies the absence of obstacles or targets within their corresponding visual occlusion zones, implying that the random selected target t is unobstructed by either obstacles or other targets. In accordance with the system model in **section 3.4**, it is obvious that both obstacles and targets are uniformly distributed and independent of each other. Furthermore, the values

of the random variables Y_{ZO_i} and Y_{ZT_i} are non-negative integers. Therefore, the probability of the event $\{Y = 0\}$ occurring can be expressed as:

$$\begin{aligned}
 P(Y = 0) &= P\left(\sum_{i=1}^w Y_{ZO_i} + \sum_{i=1}^v Y_{ZT_i} = 0\right) \\
 &= P\left(\sum_{i=1}^w Y_{ZO_i} = 0\right) \times P\left(\sum_{i=1}^v Y_{ZT_i} = 0\right) \\
 &= \bigcap_{i=1}^w P\left(Y_{ZO_i} = 0\right) \times \bigcap_{i=1}^v P\left(Y_{ZT_i} = 0\right) \\
 &= e^{-\sum_{i=1}^w \eta_i E(ZO_i)} \times e^{-\sum_{i=1}^v \mu_i E(ZT_i)} \\
 &= e^{-\left(\sum_{i=1}^w \eta_i E(ZO_i) + \sum_{i=1}^v \mu_i E(ZT_i)\right)}
 \end{aligned} \tag{13}$$

Proposition 1: For the sensor s and the target t depicted in Fig.11, the expected value $E(r)$ of the probability $P(Y = 0)$, which describes the event $\{Y = 0\}$ more accurately, can be calculated using the following formula:

$$E(r) = \frac{\int_0^r P(Y = 0) dx}{r} \tag{14}$$

Proof: According to Eq.9 10 11 12, we find that the probability $P(Y = 0)$ is related to the variable $x \in [0, r]$. To describe this probability more accurately, and given that x follows a uniform distribution, it is clear that the expected value of $P(Y = 0)$ can be calculated by integrating it over the range $[0, r]$ and then dividing the result by r . Thus, the **Proposition 1** is proven.

Base on the aforementioned derivation, for sensors in $Type_i$ and $Type_j$, if $i \neq j$, we obtain $E(r_i) \neq E(r_j)$. Substituting r_i or r_j into Eq.14 yields their respective values.

5.2. K-ITCO Estimation Model

In the previous section, we focus on deducing the probability that a random selected target t is not obstructed by obstacles and other targets, while neglecting the probability that the target is covered by at least 1 sensor. In this section, the K -ITCO estimation model is derived.

Let A_i be a random variable representing whether the target is covered by exactly one sensor in $Type_i$, where $i \in \{1, 2, \dots, u\}$. The value of A_i is either 1 (indicating coverage) or 0 (indicating no coverage). The event $\{A_i = 1\}$ denotes that the target is covered by exactly one sensor. According to the conditions 1 and 2 in **section 3.5.4**, the probability of this event can be expressed as:

$$P(A_i = 1) = \frac{E(\Phi_i)}{\Omega - \sum_{i=1}^w W_i o_i - \sum_{i=1}^v M_i t_i} \frac{\theta_i}{2\pi} \tag{15}$$

Where o_i denotes the area of an obstacle from GO_i , and t_i denotes the area of a target from GT_i . The term $\frac{E(\Phi_i)}{\Omega - \sum_{i=1}^w W_i o_i - \sum_{i=1}^v M_i t_i}$ indicates the probability that the target is located within the expected EDR of the sensor. The expression $\frac{\theta_i}{2\pi}$ represents the probability that the target lies in the sensor's FoV angle.

Subsequently, we use a random variable B_i to indicate whether the target is ITCO-covered by exactly one sensor in $Type_i$, its value is either 1 or 0. Based on the **Proposition 1** and Eq.15, it can be easily concluded that the probability of the event $\{B_i = 1\}$ occurring is given by:

$$\begin{aligned}
 P(B_i = 1) &= P(A_i)E(r_i) \\
 &= \frac{\frac{E(\Phi_i)\theta_i E(r_i)}{2\pi}}{\Omega - \sum_{i=1}^w W_i o_i - \sum_{i=1}^v M_i t_i}
 \end{aligned} \tag{16}$$

Now, a random experiment for the event $\{B_i = 1\}$ is conducted for N_i trials. Let a random variable C_i denote the number of sensors in $Type_i$ that ITCO-cover the target. Obviously, the event $\{C_i = k_i\}$ follows a Binomial Distribution, and its probability can be expressed as:

$$P(C_i = k_i) = \binom{N_i}{k_i} (P(B_i = 1))^{k_i} (1 - P(B_i = 1))^{N_i - k_i} \quad (17)$$

Using a similar proof method in **Lemma 1**, it is evident that the probability $P(C_i = k_i)$ can be approximated by a Poisson Distribution with strength $\frac{\lambda_i E(\Phi_i) \theta_i E(r_i)}{2\pi}$, where $\lambda_i = \frac{N_i}{\Omega - \sum_{i=1}^W W_i o_i - \sum_{i=1}^V M_i t_i}$. Thus, it can be expressed as:

$$P(C_i = k_i) \approx e^{-\frac{\lambda_i E(\Phi_i) \theta_i E(r_i)}{2\pi}} \frac{\left(\frac{\lambda_i E(\Phi_i) \theta_i E(r_i)}{2\pi} \right)^{k_i}}{k_i!} \quad (18)$$

Afterwards, let D be a random variable representing the total number of sensors from all types that exactly ICTO-cover the target. Clearly, the event $\{D = k\}$ indicates the target is ICTO-covered by exactly k sensors, where $k = k_1 + k_2 + \dots + k_u$, and k_i denotes the number of sensors in $Type_i$ exactly ICTO-cover the target. Therefore, the probability of the event $\{D = k\}$ is given by:

$$P(D = k) = \sum_{k=k_1+\dots+k_u} \left(\bigcap_{i=1}^u P(C_i = k_i) \right) \quad (19)$$

In accordance with **Lemma 2** in literature [19], the following expression can be derived.

$$P(D = k) \approx e^{-\frac{\sum_{i=1}^u \lambda_i E(\Phi_i) \theta_i E(r_i)}{2\pi}} \frac{\left(\frac{\sum_{i=1}^u \lambda_i E(\Phi_i) \theta_i E(r_i)}{2\pi} \right)^k}{k!} \quad (20)$$

Theorem 1: Let $\mathcal{P}(K)$ denote the K -ITCO ratio. Thus, the K -ITCO estimation model, referred to as K -ITCO with boundary effects, can be expressed using the following formula:

$$\begin{aligned} \mathcal{P}(K) &= P(D \geq K) \\ &\approx 1 - \sum_{k=0}^{K-1} \left(e^{-\frac{\sum_{i=1}^u \lambda_i E(\Phi_i) \theta_i E(r_i)}{2\pi}} \frac{\left(\frac{\sum_{i=1}^u \lambda_i E(\Phi_i) \theta_i E(r_i)}{2\pi} \right)^k}{k!} \right) \end{aligned} \quad (21)$$

Given an irregular RoI, where obstacles and irregular targets are randomly and independently located, the goal is to achieve a specified K -ITCO ratio $\mathcal{P}(K) \in [0, 1]$ with a high probability. By employing an exhaustive search method to solve Eq.21 in **Theorem 1**, the number of sensors required for random deployment can be estimated. Additionally, if the parameters related to sensors, obstacles and targets are known, the K -ITCO ratio can be estimated using Eq.21. The pseudocode for estimating the K -ITCO ratio with boundary effects is outlined in **Alg.1**.

Proposition 2: Assuming the absence of obstacles in the RoI, the K -ITCO estimation model can be expressed as:

$$\mathcal{P}(K) \approx 1 - \sum_{k=0}^{K-1} \left(e^{-\frac{\sum_{i=1}^u \check{\lambda}_i E(\Phi_i) \theta_i \check{E}(r_i)}{2\pi}} \frac{\left(\frac{\sum_{i=1}^u \check{\lambda}_i E(\Phi_i) \theta_i \check{E}(r_i)}{2\pi} \right)^k}{k!} \right) \quad (22)$$

Where $\check{\lambda}_i = \frac{N_i}{\Omega - \sum_{i=1}^V M_i t_i}$, and $\check{E}(r_i) = \frac{\int_0^{r_i} \ddot{P}(Y=0) dx}{r_i}$. Besides, $\ddot{P}(Y = 0) = e^{-\sum_{i=1}^V \ddot{\mu}_i E(ZT_i)}$.

Algorithm 1 Estimation of K -ITCO ratio with boundary effects (K-ITCO-BE)

Input: 1) Parameters of sensors: u, N_i, r_i and θ_i ; 2) Parameters of obstacles: $w, o_i, \mathbf{ov}_i, W_i, ol_i$ and ow_i ; 3) Parameters of targets: $v, t_i, \mathbf{tv}_i, M_i, tl_i$ and tw_i ; 4) K

Output: K -ITCO ratio

```

1: for each  $GO_i$  obstacle do
2:   Compute its density  $\eta_i$ , and  $P(Y_{ZO_i} = 0)$  by Eq.11
3: end for
4: for each  $GT_i$  target do
5:   Compute its density  $\mu_i$ , and  $P(Y_{ZT_i} = 0)$  by Eq.12
6: end for
7: Compute  $P(Y = 0)$  by Eq.13
8: for each  $Type_i$  sensor do
9:   Compute its density  $\lambda_i$ 
10:  Calculate  $E(r_i)$  using Eq.14, and  $E(\Phi_i)$  using Eq.1.
11: end for
12: Substitute  $\lambda_i, E(\Phi_i), E(r_i), \theta_i, u, K$  and other parameters into Eq.21, then we get  $P(K)$ 
13: return  $P(K)$ 

```

Proof: In the absence of obstacles, it is clear that the visual occlusion zone formed by obstacle does not exist. According to the derivation of Eq.11, we obtain $P(Y_{ZO_i} = 0) = 1$. By substituting it into Eq.13, we can derive the formula for $\ddot{P}(Y = 0)$. Subsequently, we compute $\ddot{E}(r_i)$ by substituting $\ddot{P}(Y = 0)$ into Eq.14. Using the equation in **Theorem 1**, the **Proposition 2** can be easily derived.

Proposition 3: Ignoring the boundary effects, the K -ITCO estimation model, referred to as K -ITCO without boundary effects, can be expressed as:

$$P(K) \approx 1 - \sum_{k=0}^{K-1} \left(e^{\frac{-\sum_{i=1}^u \lambda_i r_i^2 \theta_i E(r_i)}{2}} \frac{\left(\frac{\sum_{i=1}^u \lambda_i r_i^2 \theta_i E(r_i)}{2} \right)^k}{k!} \right) \quad (23)$$

Proof: If we ignore the boundary effects, based on the derivation of $E(\Phi_i)$, we get $E(\Phi_i) = \pi r_i^2$. By substituting it into Eq.21, **Proposition 3** can be easily proven.

The pseudocode for estimating the K -ITCO ratio without boundary effects is outlined in **Alg.2**.

Proposition 4: Given a homogeneous WWSN in the RoI where only one type of sensors is randomly deployed, it can be readily concluded that the K -ITCO estimation model can be expressed by the following formula:

$$P(K) \approx 1 - \sum_{k=0}^{K-1} \left(e^{\frac{-\lambda E(\Phi) \theta E(r)}{2\pi}} \frac{\left(\frac{\lambda E(\Phi) \theta E(r)}{2\pi} \right)^k}{k!} \right) \quad (24)$$

Proof: In accordance with the deduction of **Theorem 1**, the **Proposition 4** can be easily proven.

Now, we will provide an example to verify the accuracy of our proposed K -ITCO estimation model in Eq.21.

Example. As depicted in Fig.12, consider a square RoI Ω with side length of 100m, where 20 homogeneous irregular targets and 8 obstacles are randomly and independently distributed. The areas of targets and obstacles are specified as 3.7213m² and 99.7599m², respectively. According to the target and obstacle models, the minimum bounding rectangles for both target and obstacle have dimensions of $tl = 1.5440m$, $tw = 3.2274m$ and $ol = 11.3649m$, $ow = 10.7968m$, respectively. Furthermore, the relevant parameters for sensors in $Type_1$ and $Type_2$ are set as $r_1 = 10m$, $\theta_1 = \frac{2\pi}{3}$ and $r_2 = 15m$, $\theta_2 = \frac{\pi}{3}$. To achieve $P(1) \geq 90\%$ with a high probability and reduce the complexity of

Algorithm 2 Estimation of K -ITCO ratio no boundary effects (K-ITCO-noBE)

Input: 1) Parameters of sensors: u , N_i , r_i and θ_i ; 2) Parameters of obstacles: w , o_i , $\mathbf{ov}_i$, W_i, ol_i and ow_i ; 3) Parameters of targets: v , t_i , $\mathbf{tv}_i$, M_i, tl_i and tw_i ; 4) K

Output: K -ITCO ratio

```

1: for each  $GO_i$  obstacle do
2:   Compute its density  $\mu_i$ , and  $P(Y_{ZO_i} = 0)$  by Eq.11
3: end for
4: for each  $GT_i$  target do
5:   Compute its density  $\mu_i$ , and  $P(Y_{ZT_i} = 0)$  by Eq.12
6: end for
7: Compute  $P(Y = 0)$  by Eq.13
8: for each  $Type_i$  sensor do
9:   Compute corresponding density  $\lambda_i$ 
10:  Calculate  $E(r_i)$  using Eq.14
11: end for
12: Substitute  $E(r_i)$ ,  $\lambda_i$ ,  $r_i$ ,  $\theta_i$ ,  $u$ ,  $K$  and other parameters into Eq.23, we obtain  $\mathcal{P}(K)$ 
13: return  $\mathcal{P}(K)$ 

```

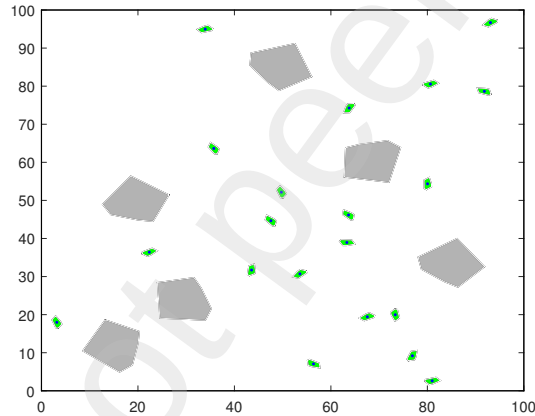


Figure 12: Illustration of system model in example where 6 obstacles and 20 targets are located within the square RoI. The obstacles are depicted as gray convex polygons, while the targets are represented by green convex polygons.

the exhausting search method, we assume that $N_1 = N_2$. According to the **Alg 1**, we obtain that $E(\Phi)_1 = 285.8696m^2$, $E(\Phi)_2 = 616.6849m^2$, and $E(r_1) = 83.1\%$, $E(r_2) = 80.07\%$. Substituting these parameters into Eq.21 as outlined in **Theorem 1** using exhausting search method, it is determined that $\lambda_1 = \lambda_2 = 142 \times 10^{-4}$. Consequently, we find that $N_1 = N_2 = 133$. This indicates that at least 133 sensors of $Type_1$ and 133 sensors of $Type_2$ needed to be deployed in the RoI to ensure that over 90% targets are 1-ITCO covered with a high probability.

6. Simulation Experiments

In this section, a series of simulation experiments using Matlab 2018a are respectively conducted to verify the accuracy of numerical results obtained from Eq.21 in **Theorem 1** and Eq.23 in **Proposition 3**. Our primary focus is on investigating the influence of three metrics—sensor density, sensing radius and Fov angle—on the numerical and simulation results of K -ITCO ratio.

As illustrated in Fig.13 It is assumed that the RoI is a convex irregular region containing 15 targets from GT_1 and 15 targets from GT_2 , as well as randomly positioned 5 obstacles from GO_1 and 5 obstacles from GO_2 , which are independent of each other. In addition, we assume that the shapes and areas of the RoI, targets and obstacles are fixed.

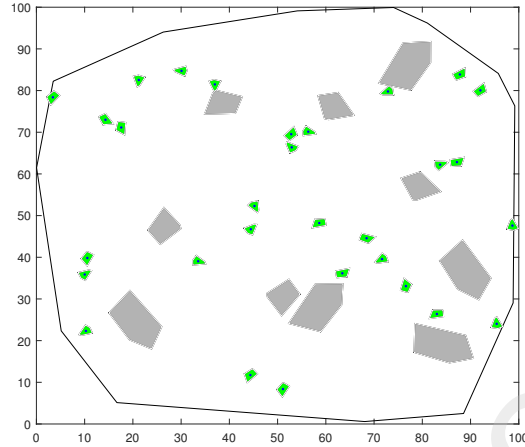


Figure 13: Illustration of system model in simulation experiments. The obstacles are depicted as gray convex shapes, while the targets are represented by green convex shapes. The RoI is depicted as the gray-lined convex shape.

Table 2
Main Parameters in Simulations

Parameter	Description	Values
Ω	Area of RoI	$8483.7m^2$
W_1	Number of obstacles in GO_1	5
W_2	Number of obstacles in GO_2	5
o_1	Area of GO_1 obstacle	$31.5075m^2$
o_2	Area of GO_2 obstacle	$80.0521m^2$
M_1	Number of targets in GT_1	15
M_2	Number of targets in GT_2	15
t_1	Area of GT_1 target	$3.8090m^2$
t_2	Area of GT_2 target	$4.6035m^2$

The area of RoI is specified as $8483.7m^2$, the areas of targets in GT_1 and GT_2 are set as $3.809m^2$ and $4.6035m^2$, and the areas of obstacles in GO_1 and GO_2 are calculated to be $31.5075m^2$ and $80.0521m^2$. For simplifying the simulation experiments, we only take two types of sensors ($Type_1$ and $Type_2$) into consideration and assume that scale of each type is identical. Thus, we get $N_1 = N_2$. The main parameters used in experiments are summarized in Table 2.

Subsequently, to obtain more accurately simulation results, each set of simulation experiments is conducted 100 times. The final results of each set are presented as the mean values to ensure statistical robustness.

6.1. Influence of Sensor Density

In this simulation, we primarily analyze the influence of sensor density on the numerical and simulation results of K -ITCO, and relevant parameters of sensors are set as follows: for $Type_1$ sensors, $r_1 = 15m$ and $\theta_1 = \frac{\pi}{2}$; for $Type_2$ sensors, $r_2 = 10m$ and $\theta_2 = \frac{2\pi}{3}$. Additionally, the densities (λ_1 and λ_2) for these sensors are both varied from 20×10^{-4} to 120×10^{-4} in increments of 20×10^{-4} .

As illustrated in Fig.14, the trends imply that the K -ITCO ratio rises as the sensor density increases, this is because a higher density indicates that more sensors can contribute to the coverage. It is noteworthy that the K -ITCO ratio approaches to 0 at low densities and approaches to 1 at high densities, indicating that the boundary effects are not prominent at these extremes. This can be attributed to the fact that a lower density results in fewer sensors falling into the boundary zone, while a higher density leads to K -ITCO ratio of 1, thus diminishing the impact of boundary effects. Additionally, for a constant density, we observe that the K -ITCO ratio increase as K decreases. This occurs because a lower K requires fewer sensors to meet the coverage requirements, thereby facilitating a higher K -ITCO ratio. As a

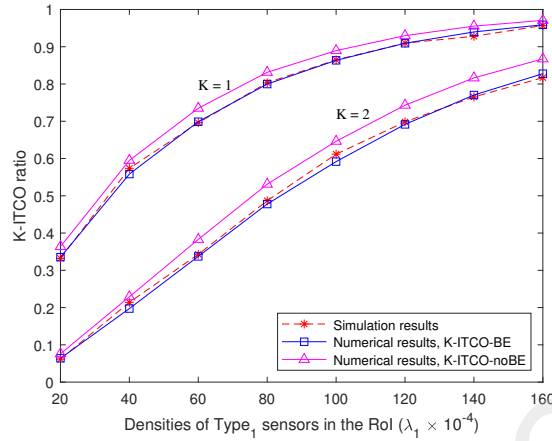


Figure 14: Influence of sensor density on K -ITCO ratio, where $\lambda_1 = \lambda_2$.

whole, the numerical results accounting for boundary effects (K -ITCO-BE) demonstrate a closer alignment with the simulation results compared to those without boundary effects (K -ITCO-noBE).

6.2. Influence of Sensing Radius

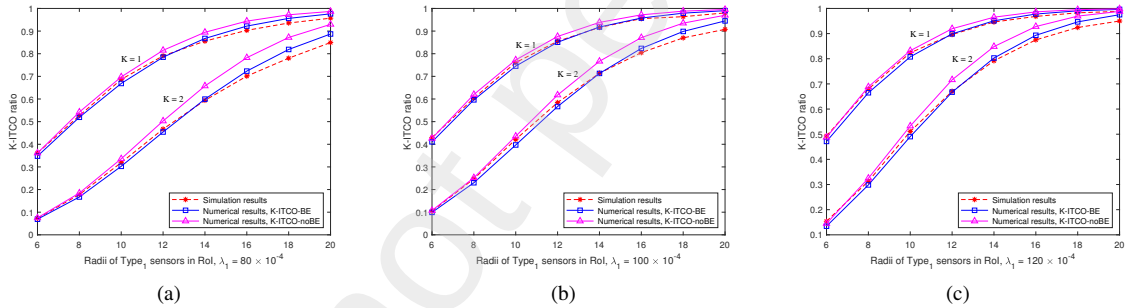


Figure 15: Influence of sensing radius on K -ITCO ratio, where $\theta_1 = \theta_2$ and $\lambda_1 = \lambda_2$. (a) $\lambda_1 = 80 \times 10^{-4}/m^2$, (b) $\lambda_1 = 100 \times 10^{-4}/m^2$, (c) $\lambda_1 = 120 \times 10^{-4}/m^2$.

In this simulation, we focus on the influence of sensing radius on the numerical and simulation results of K -ITCO. The relevant parameters for $Type_1$ and $Type_2$ sensors are configured as follows: $\lambda_1 = \lambda_2$, $\theta_1 = \frac{\pi}{2}$ and $\theta_2 = \frac{2\pi}{3}$. The sensing radii (r_1 and r_2) are initially set to $6m$ and incremented by $2m$ up to maximum $20m$.

As shown in Fig.15, the numerical results obtained by our models (**Theorem 1** and **Proposition 3**) demonstrate good performance in fitting the simulation results, and it is evident that the K -ITCO-BE has better accuracy than the K -ITCO-noBE. Intuitively, we observe that an increase in the sensing radius leads to a rise in the K -ITCO ratio, regardless of the values of K and the sensor densities (80×10^{-4} , 100×10^{-4} and 120×10^{-4}). Moreover, it is readily apparent that the impact of boundary effects is negligible when the sensing radius is small. This is because a smaller sensing radius ensures that the corresponding sensing region remains largely within the RoI, thereby diminishing the prominence of the boundary effects. Conversely, the influence of boundary effects becomes more pronounced when the sensing radius is larger and the K -ITCO ratio is not close to 1. This is attributed to the fact that an expanded sensing radius enlarges the sensing region, resulting in a greater portion of the sensing region fall outside of the RoI. Notably, as the sensing radius increases to a substantial value, the K -ITCO ratio approaches 1 and the boundary effects gradually diminish. Furthermore, for a fixed sensing radius, it is evident that K -ITCO ratio increases as K decreases. This is

due to the fact that a lower value of K necessitates a smaller sensing radius to satisfy the coverage criteria, thereby facilitating a higher K -ITCO ratio.

6.3. Influence of FoV

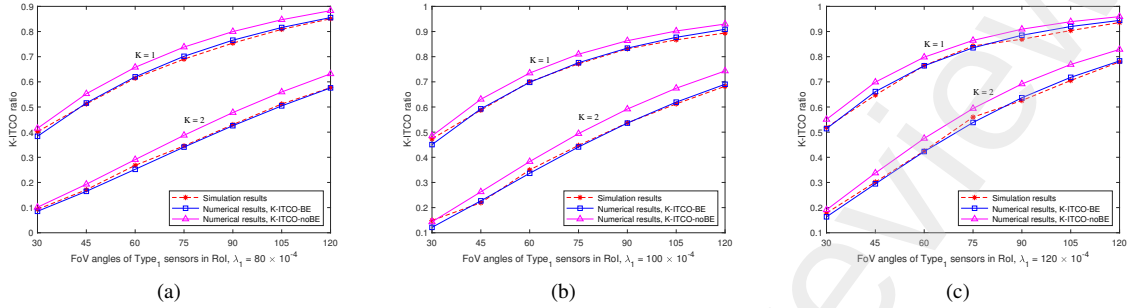


Figure 16: Influence of FoV angle on K -ITCO ratio, where $r_1 = r_2$ and $\lambda_1 = \lambda_2$. (a) $\lambda_1 = 80 \times 10^{-4}/m^2$, (b) $\lambda_1 = 100 \times 10^{-4}/m^2$, (c) $\lambda_1 = 120 \times 10^{-4}/m^2$.

In this simulation, we consider the influence of FoV on the numerical and simulation results of K -ITCO. The relevant parameters of sensors in $Type_1$ and $Type_2$ are respectively configured as follows: $\lambda_1 = \lambda_2$, $r_1 = 15m$ and $r_2 = 10m$. Additionally, the FoV angles (θ_1 and θ_2) are both varied from $\frac{\pi}{3}$ to $\frac{2\pi}{3}$ in increments of $\frac{\pi}{6}$.

As depicted in Fig.16, the trends reveal that accuracy of numerical results calculated through **Alg.1** and **Alg.2** is satisfactory, and it is obvious that the K -ITCO-BE demonstrates superior performance compared to the K -ITCO-noBE. Empirically, the K -ITCO ratio consistently rises as the FoV expands, irrespective of the values of K and different sensor densities (80×10^{-4} , 100×10^{-4} and 120×10^{-4}). Moreover, it is observed that the influence of boundary effects remains stable. this stability is attributed to the fact that the boundary effects are primarily determined by the EDR of sensor located in the boundary zone, while the FoV does not significantly impact the EDR. Additionally, for a constant value of FoV, it is noteworthy that a decrease in K leads to an increase in the K -ITCO ratio, as a lower K necessitates a smaller FoV to fulfill the coverage requirements.

7. Conclusion

In this paper, we investigate the coverage estimation issue for irregular targets in WVSNs deployed in region of interest containing obstacles. In contrast to traditional target coverage issue that discard the target's shape and size, we introduce a novel irregular target model and obstacle model, and propose a new coverage metric termed ITCO coverage. To address this issue, we derive the expected value of the effective detection region (EDR) and the probability that a target is not obstructed. Subsequently, we further derive the K -ITCO estimation model that accounts for boundary effects (K -ITCO-BE) and propose a corresponding K -ITCO model that disregards boundary effects (K -ITCO-noBE). Finally, a series of simulations are conducted to evaluate the accuracy of numerical results obtained by our proposed models. The findings of this study provide valuable insights for engineers, enabling more effective deployment of initial WVSNs based on varying QoC requirements.

Future research will explore several critical aspects of WVSNs randomly deployed in an irregular RoI containing obstacles. First, we aim to develop strategies to enhance the K -ITCO ratio using a minimum number of sensors. Second, we intend to design an optimal charging strategy to ensure the longevity and security of the network, while simultaneously improving the K -ITCO performance.

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